

Meherun Mehnaj Miti

Data Science Undergraduate Student, United International University, Dhaka, Bangladesh
mmiti2330122@bsds.uiu.ac.bd | 01517325105 | [Portfolio](#) | [Blog Website](#) | [LinkedIn](#) | [GitHub](#)

RESEARCH INTERESTS

Data Science & Machine Learning | LLMs, Reasoning Architectures & Prompt Engineering | Agentic AI | Detection Models & Transformers

WORK EXPERIENCE

Archives and Digitization Manager – *Otibeguni* November 2024 – Present

- Digitized and extracted text from rare documents using **pytesseract**, **OpenCV**, and **Pillow**; cleaned and structured data with **pandas**, **re**, and **spaCy**; refined extracted text using the **ChatGPT API** and Python NLP tools for improved research usability.
- Applied preprocessing techniques to improve **OCR** accuracy and reduced manual correction effort by **30%**.
- Extracted and cleaned hard-to-read text from old Bangla and Sylheti Nagri documents, improving usability of rare archival materials.
- Organized cleaned data using **Airtable** to create a searchable digital archive, improving accessibility for researchers and collaborators.

Project Coordinator Intern – *Advanced Intelligent Multidisciplinary Systems Lab* October 2024 – Present
Institute of Research, Innovation, Incubation and Commercialization (IRIIC), AIMS Lab, UIU

- Supported development of an [AI-based Prescription Audit System](#) to automate error detection in medical prescriptions using ML algorithms, enhancing patient safety and clinical efficiency.
- Coordinated documentation, project paperwork, and research operations across ongoing lab initiatives.

Undergraduate Teaching Assistant (UGA) – *United International University* July 2024; October 2024 – Present

Courses:

Programming for Data Science Laboratory (DS-1502)

Object Oriented Programming for Data Science Laboratory (DS-1116)

- Assisted faculty in preparing problem sets.
- Resolved student doubts and academic queries.
- Conducted extra problem-solving sessions and provided 1-on-1 support, helping students clarify core programming concepts.

Data Science and Python Instructor, UIU Data Science Club February 2025 – Present

- Delivered Python and data science sessions with practical examples for student learners.

PROJECTS

- NeuroChain**
Developed NeuroChain, an AI-powered full-stack platform for autism assessment, gamified therapy, and specialist collaboration; 1st Runner-up at HULT Prize National Bangladesh 2026 and HULT Prize UIU 2026, National Finalist at Solvio AI Hackathon 2025, and 6th Runner-up in the DBMS Project Show.
- GraphReAct** *Ongoing*
Implemented and evaluated an AGoT-ReAct framework combining Algorithm of Thoughts and ReAct for GPQA mathematical reasoning tasks using DeepSeek and Qwen models; achieved best average method performance of **31.91%** and reduced Qwen reasoning steps by **56.8%**.
- CausalTrace-MLSD**
Built a reproducible MLSD pipeline to evaluate ROME and MEMIT knowledge edits on GPT-2 XL using a custom Causal Consistency Score across **200** prediction runs; found ROME reached **93.8%** edit success but only **17.9%** causal consistency.
- MediaMorphBD**
Created a government-stance classification dataset through collection, cleaning, and structured wrangling for downstream NLP analysis.

- **Tribal Threads**
App for Indigenous craft promotion and digital browsing.

SKILLS

- **Programming Languages:** Python, R, SQL, HTML, CSS
- **Technologies, Frameworks & Tools:** Scikit-Learn, PyTorch, Tesseract, Streamlit, FastAPI, MLflow, DVC, Feast, Git, GitHub, MySQL, PostgreSQL
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Power BI
- **Microsoft Office Suite:** Word, Excel, PowerPoint
- **Artificial Intelligence & Language Processing:** LLM APIs, prompt engineering, ReAct/CoT/GoT, knowledge editing, ML, NLP, Agentic AI, reasoning algorithms, and detection models.
- **Soft Skills:** Strong leadership, effective communication and public speaking, analytical thinking and problem-solving, time management, and creativity.

EXTRACURRICULAR ACTIVITIES

Founding Joint Secretary – UIU Data Science Club February 2025 – Present

- Contributed to decision-making, event planning, and records management for meetings and activities.
- Conducted academic sessions for club members.

Vice President (Debate and Workshop) – Debating Club of Rouf College 2022 – 2023

- Managed debate instructors and directed and evaluated new debaters.
- Arranged events, workshops, and practice debates for members.

AWARDS

- **UIU Academic Excellence Scholarship:** 2023 – Present
Received academic excellence scholarship from UIU for **8 consecutive trimesters**, including **6 full-tuition awards**.
- **HULT Prize National Bangladesh 2026 – 1st Runner-up – NeuroChain** 2026
- **HULT Prize UIU 2026 – 1st Runner-up – NeuroChain** 2026
- **SOLVIO AI Hackathon 2025 – National Finalist – NeuroChain (out of 650+ teams)** 2025
- **UIU CSE Project Show Spring 2024 – 3rd Runner-up (OOP for Data Science Lab)** 2024
- **UIU CSE Project Show Spring 2025 – 6th Runner-up (Database Management System Lab)** 2025
- **Code of Clash Programming Contest (Junior) – Top Solver (Female)** 2024

EDUCATION

United International University 2023 – Present
B.Sc. in Data Science (Current CGPA: **3.96/4.00**)

Birshrestha Munshi Abdur Rouf Public College, Dhaka 2022
Higher Secondary School Certificate (HSC), Science, GPA-5.00

YWCA Junior Girls' High School, Dhaka 2020
Secondary School Certificate (SSC), Science, GPA-5.00

REFERENCES

Dr. Swakkhar Shatabda

Professor of Computer Science and Engineering
& Former Director, Data Science Program
United International University
Professor, BRAC University
Dhaka, Bangladesh
E-mail: swakkhar.shatabda@bracu.ac.bd
[Google Scholar](#) [BRAC University – Personal Page](#)

Saddam Hossain Mukta

Post-doctoral Researcher
LUT University
E-mail: saddam.mukta@lut.fi
[Google Scholar](#)